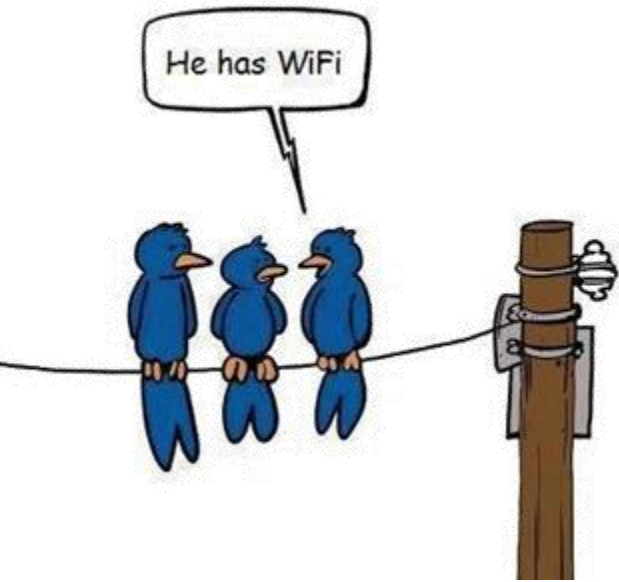


ECSE 4560/ 6560

Modern Communication Systems

Lecture 19: MIMO Part4: Multiplexing



Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Multiplexing**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Project mid-term report due **Friday 11/7 (Extension?)**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 (MIMO Theory) due **Mon 11/10**. HW6 (MIMO code) due 11/14
 - Exam 2: **Friday Nov 21**

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**

Today's Class

- MIMO Gains
 - Receiver Diversity
 - Transmitter Diversity
 - Space-Time codes (Alamouti Codes)
- MIMO Multiplexing Gains
 - **Zero-forcing**
 - **SVD**
 - **MU MIMO**
- Clock Synchronization in MIMO

MIMO Gains

Multiplexing Gain:

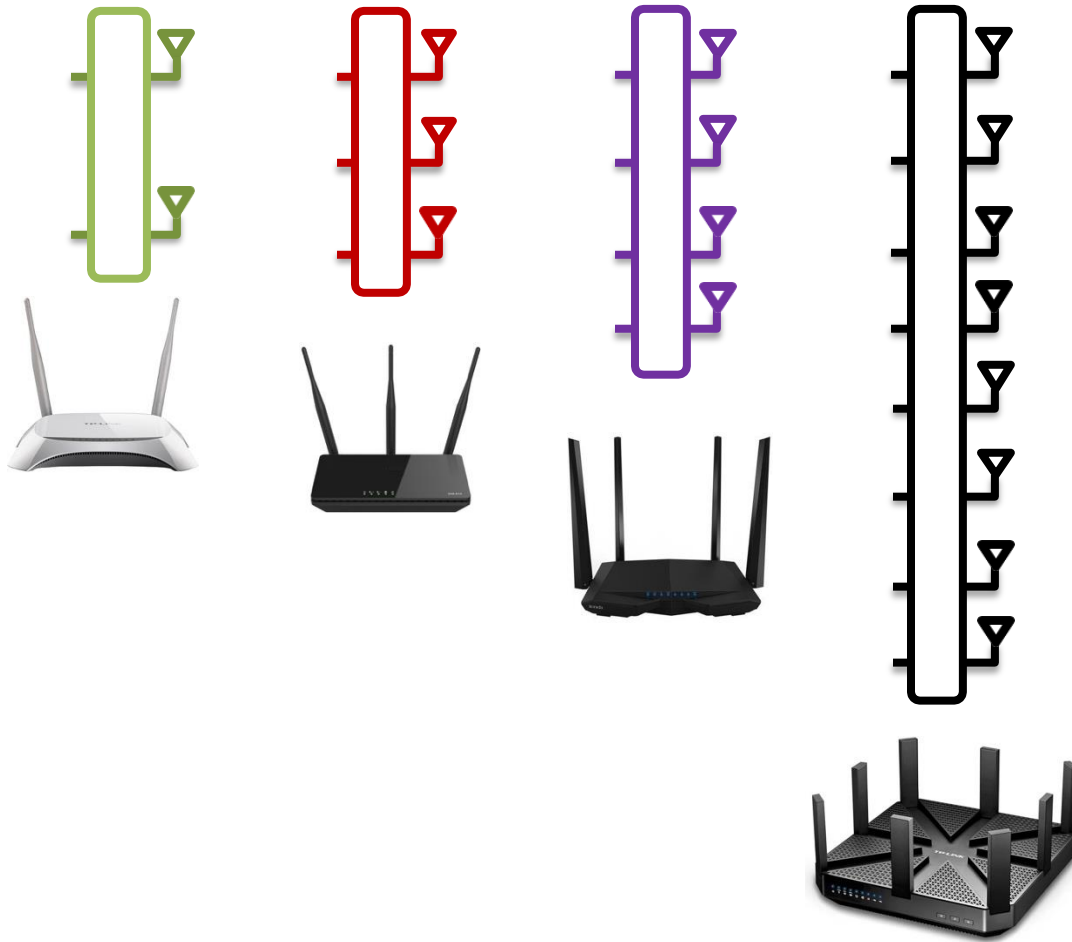
- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow \times N$ more packets
- Data Rate: $\propto N \log(SNR/N)$

Diversity Gain:

- Increase SNR of the received packets
- $N \times 1$ or $1 \times N$ MIMO $\rightarrow \times \log N$ rate
- $N \times N$ MIMO $\rightarrow \times \log N^2$ data rate
- Data Rate $\propto \log(SNR \times N^2)$

MIMO: Multiple TX-RX streams

WiFi Access Points 2 to 8 MIMO

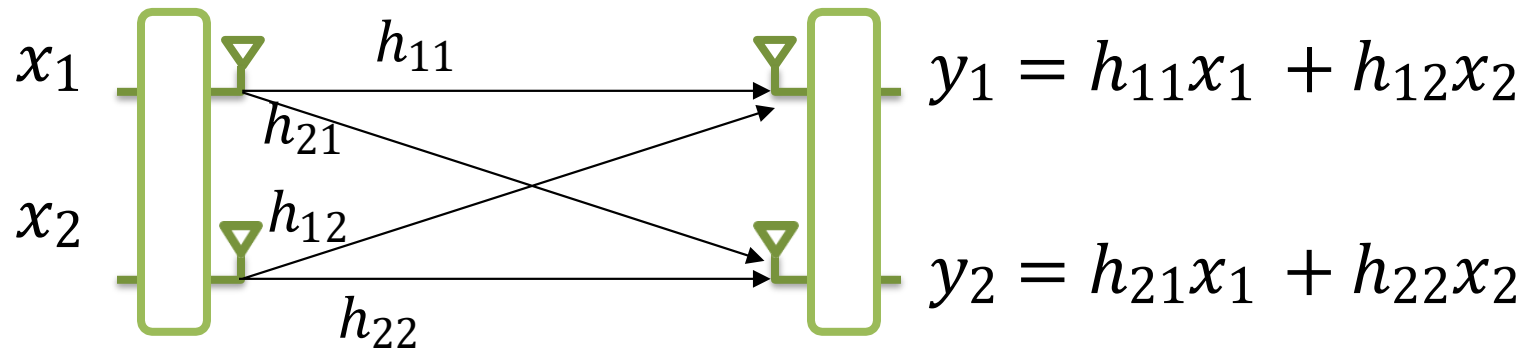


WiFi Devices 1 to 2 MIMO



- Power
- Form Factor & Antenna Separation
- Most mobile phones support 2 MIMO

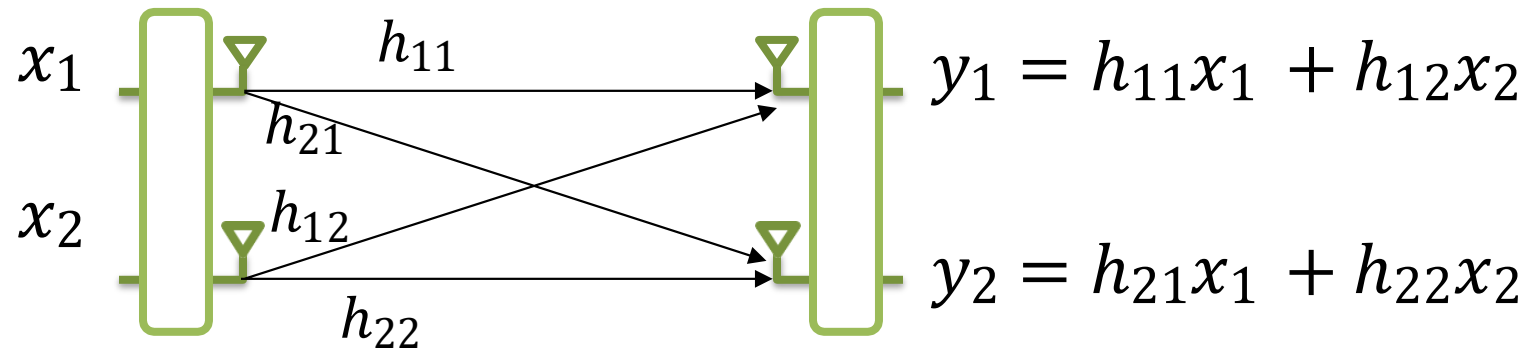
MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

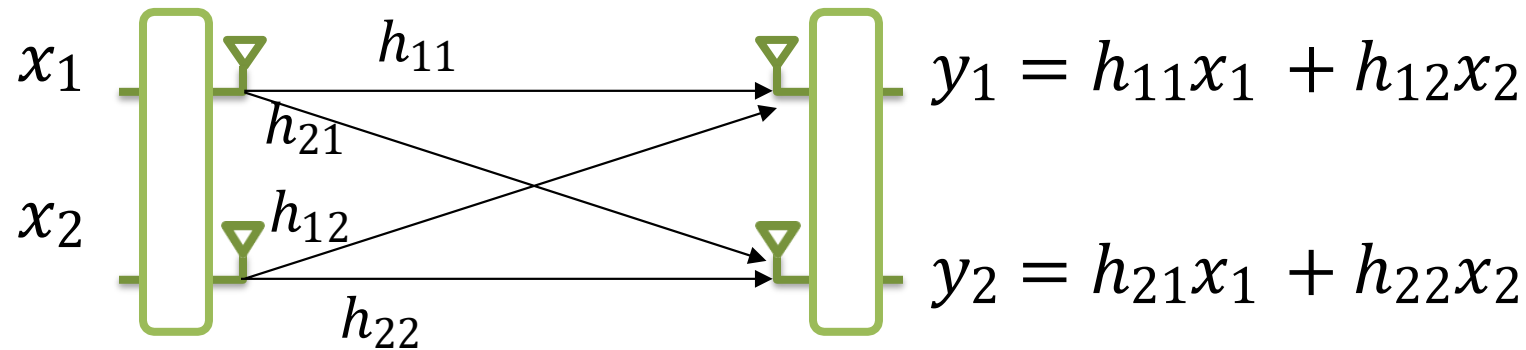
$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Zero-forcing based MIMO



$$\hat{\mathbf{x}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

Zero-forcing based MIMO



Pros

- Very simple, low latency; no transmitter CSI required.
- Completely removes inter-stream interference when $\mathbf{H}$ is well-conditioned.
- Works symbol-by-symbol; easy to pair with standard demappers.

Cons

- **Noise amplification** when $\mathbf{H}$ is ill-conditioned (small singular values).
- Sensitive to channel-estimation error.
- Not capacity-optimal; BER can be far from MMSE/ML at low–mid SNR.
- Fails if $\mathbf{H}$ is rank-deficient (needs regularization).

Question 3: Simple 2×2 MIMO with Zero-Forcing

Setup. Consider a narrowband flat-fading 2×2 MIMO link for a single channel use:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad \mathbf{H} \in \mathbb{C}^{2 \times 2} \text{ invertible,}$$

with $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\mathbb{E}[|x_1|^2] = \mathbb{E}[|x_2|^2] = P$, and $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_2)$. The receiver knows $\mathbf{H}$ perfectly.

- (a) **ZF equalizer and effective noise (one-step derivation).** Use the zero-forcing equalizer $\mathbf{W} = \mathbf{H}^{-1}$. Show that

$$\hat{\mathbf{x}} = \mathbf{W}\mathbf{y} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}.$$

Write the two decision statistics explicitly:

$$\hat{x}_1 = [\mathbf{H}^{-1}]_{1,:} \mathbf{y}, \quad \hat{x}_2 = [\mathbf{H}^{-1}]_{2,:} \mathbf{y},$$

and identify the corresponding effective noise terms on each stream.

- (b) **Numeric evaluation.** Let

$$\mathbf{H} = \begin{bmatrix} 1 & 0.5 \\ -0.3 & 1.2 \end{bmatrix}, \quad P = 1, \quad N_0 = 0.2.$$

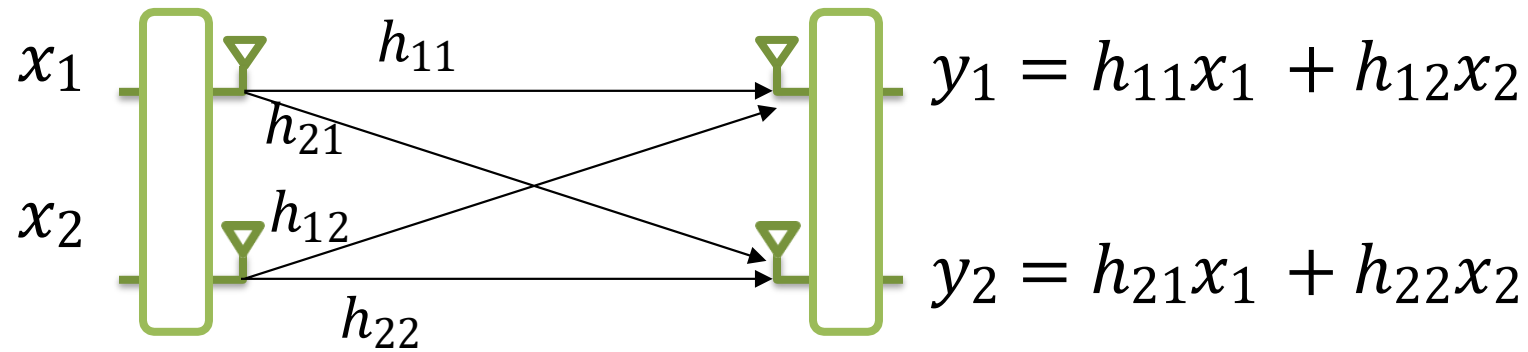
Compute:

- (i) $\mathbf{H}^{-1}$ (show your intermediate steps briefly).
- (ii) The squared row norms of $\mathbf{H}^{-1}$: $\|\mathbf{w}_1\|^2 = \|[\mathbf{H}^{-1}]_{1,:}\|^2$, $\|\mathbf{w}_2\|^2 = \|[\mathbf{H}^{-1}]_{2,:}\|^2$.
- (iii) The post-ZF SNRs using these norms (no extra matrices):

$$\gamma_1 = \frac{P}{N_0 \|\mathbf{w}_1\|^2}, \quad \gamma_2 = \frac{P}{N_0 \|\mathbf{w}_2\|^2}.$$

- (iv) Which stream is more noise-enhanced? State your answer and the SNR ratio γ_1/γ_2 in dB.

SVD-based MIMO



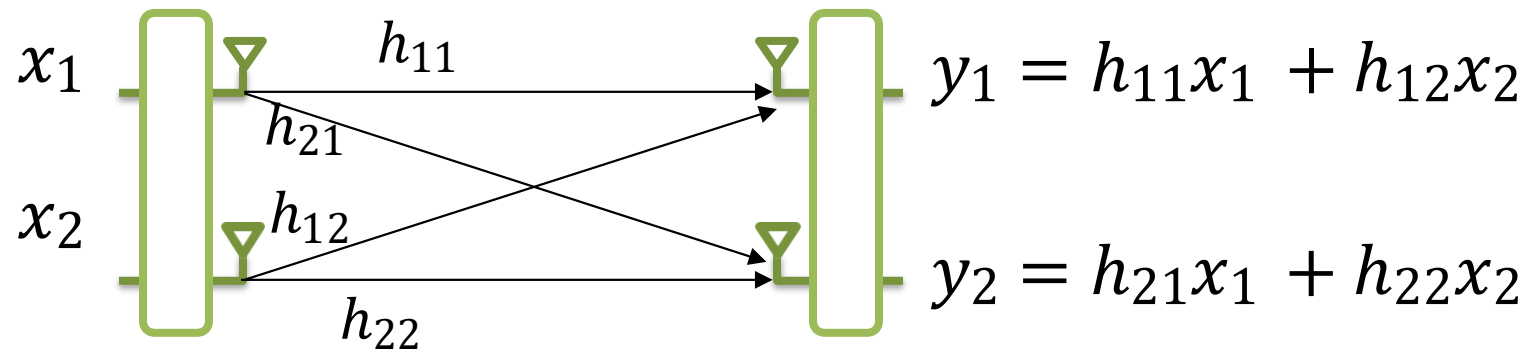
$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{array}{l} \vdots \\ \vdots \end{array} \right\} \mathbf{v}_1^H$
 $\left. \begin{array}{l} \vdots \\ \vdots \end{array} \right\} \mathbf{v}_2^H$
 $\left. \vdots \right\} \mathbf{v}_t^H$

$$= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\|\mathbf{u}_i\|^2 = \|\mathbf{v}_i\|^2 = 1, 1 \leq i \leq t$$

$$\mathbf{u}_i^H \mathbf{u}_j = \mathbf{v}_i^H \mathbf{v}_j = 0, i \neq j, 1 \leq i, j \leq t$$

$$\mathbf{V}^H \mathbf{V} = \mathbf{V} \mathbf{V}^H = \mathbf{I}_t$$

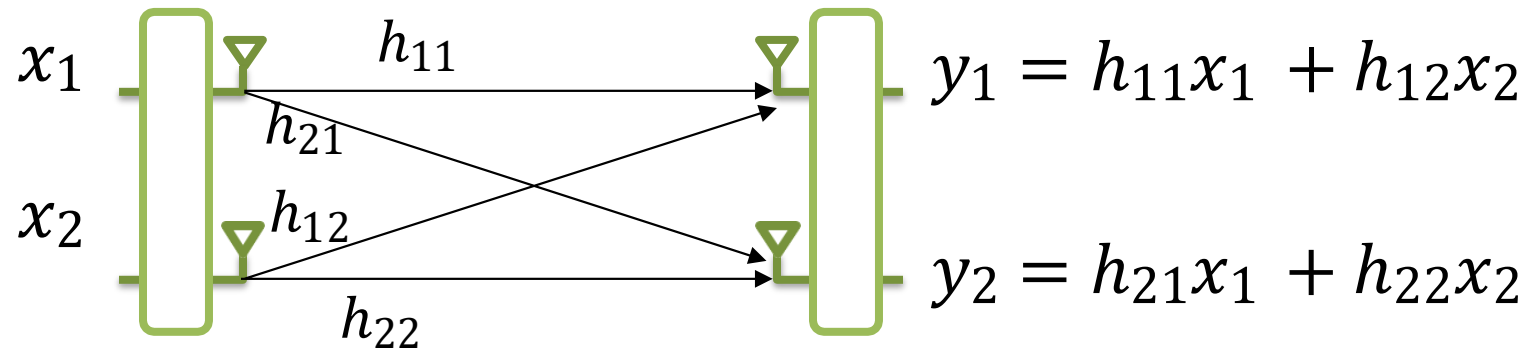
$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_t$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{matrix} \vdots \\ \vdots \end{matrix} \right\} \mathbf{v}_1^H$
 $\left. \begin{matrix} \vdots \\ \vdots \end{matrix} \right\} \mathbf{v}_2^H$
 $\left. \begin{matrix} \vdots \\ \vdots \end{matrix} \right\} \mathbf{v}_t^H$

$$= \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{y}} = \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

$$\mathbf{y} = \underbrace{\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H}_{\mathbf{H}}\mathbf{x} + \mathbf{n}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{V}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

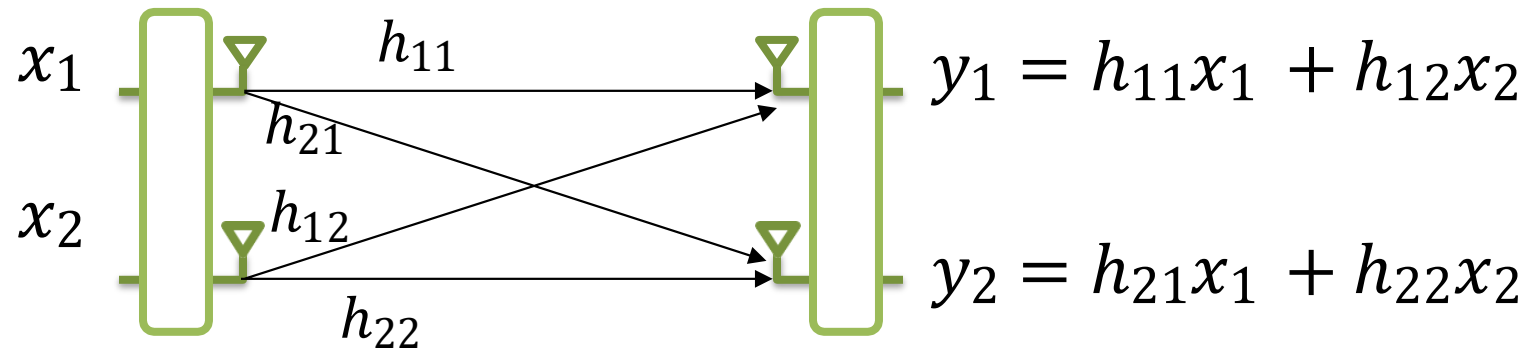
$$\underbrace{\mathbf{U}^H\mathbf{y}}_{\tilde{\mathbf{y}}} = \mathbf{U}^H (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \mathbf{n})$$

$$= \mathbf{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

$$\tilde{\mathbf{y}} = \underbrace{\mathbf{U}^H\mathbf{U}}_{\mathbf{I}_t}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \underbrace{\mathbf{U}^H\mathbf{n}}_{\tilde{\mathbf{n}}}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

SVD-based MIMO



$$\begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \vdots \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_t \end{bmatrix} + \begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \vdots \\ \tilde{n}_t \end{bmatrix}$$

$$\tilde{y}_1 = \sigma_1 \tilde{x}_1 + \tilde{n}_1 \qquad C_i = \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

$$\tilde{y}_2 = \sigma_2 \tilde{x}_2 + \tilde{n}_2$$

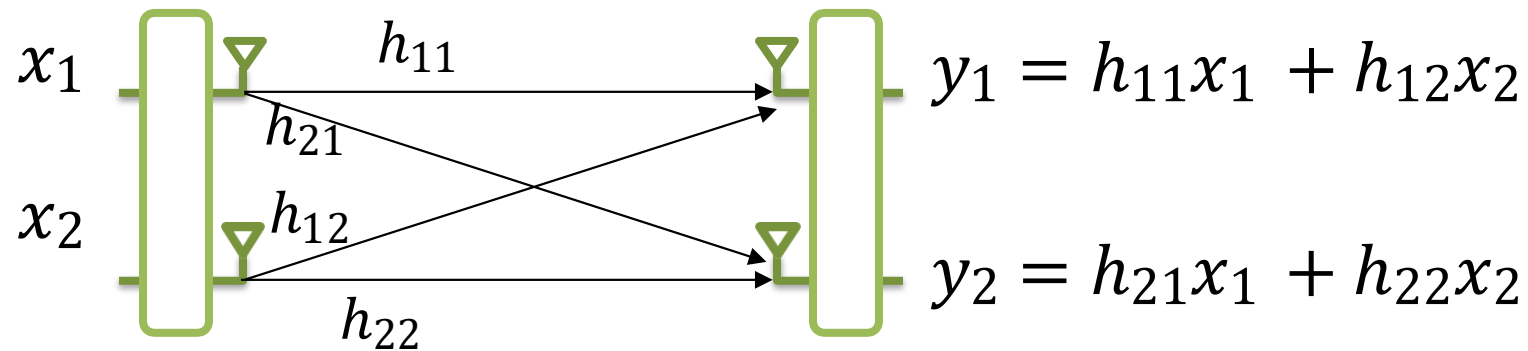
$\vdots$

$$\tilde{y}_t = \sigma_t \tilde{x}_t + \tilde{n}_t$$

Total Capacity

$$C = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_i^2} \right)$$

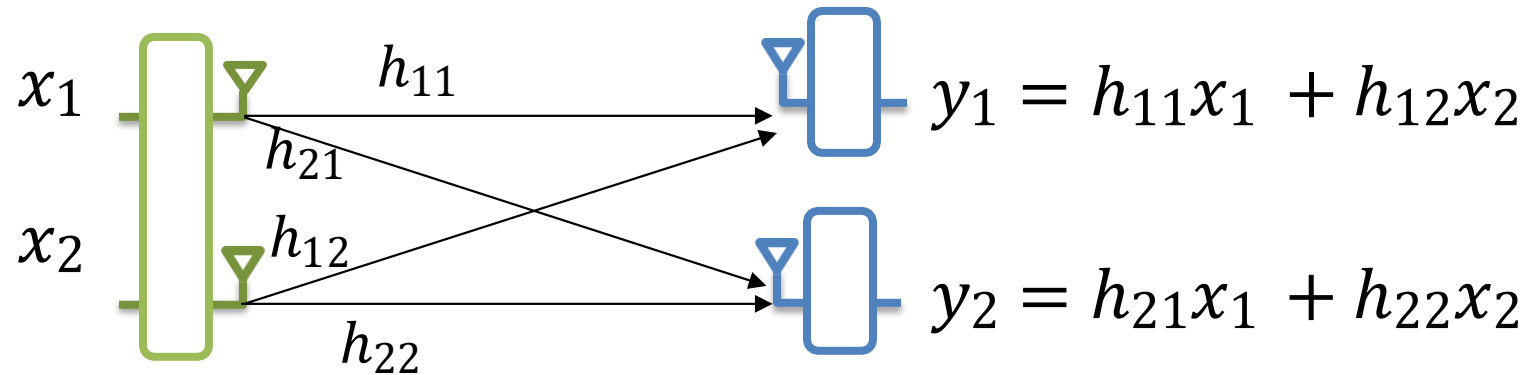
SVD-based MIMO: Power allocation?



Total Capacity

$$C = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_i^2} \right)$$

Multi-User MIMO

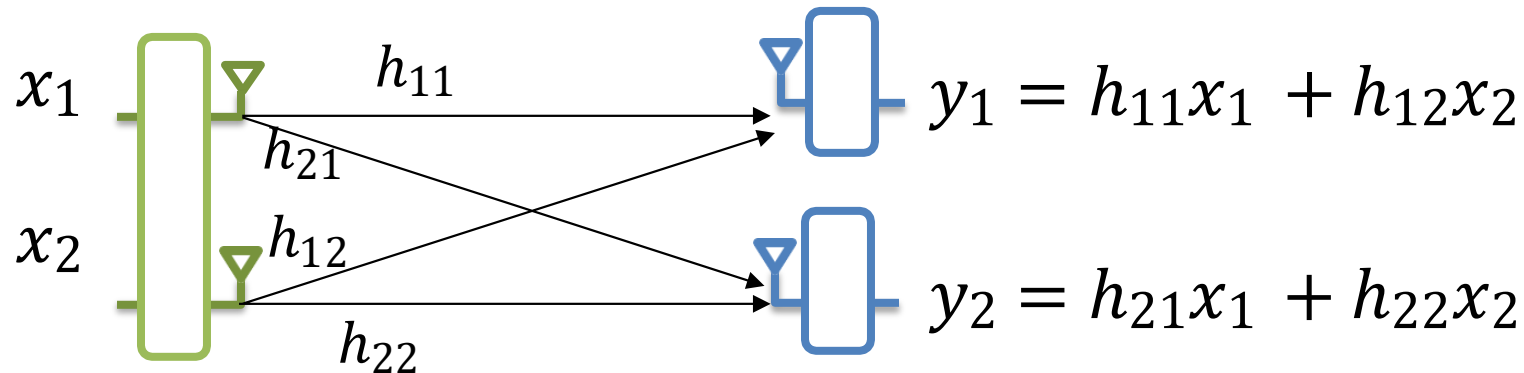


$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Does not work. Receivers do not have access to each other's signals

Precoding



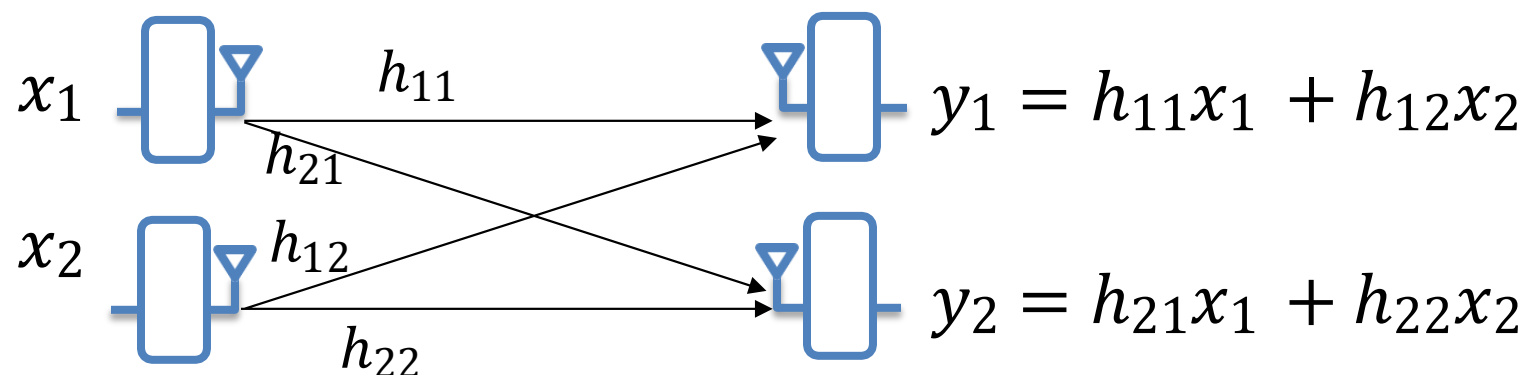
Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

Also known as Beamforming or Zero-Forcing

Requires Channel Feedback!

What about distributed transmitters?



Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

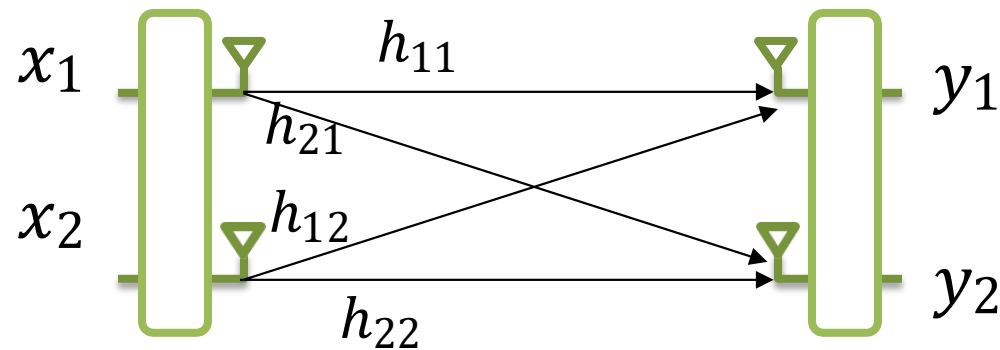
Does not work. Transmitters do not have access to each other's signals

False: what if transmitters are APs connected over Ethernet?

Today's Class

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 - Zero-forcing
 - SVD
 - MU MIMO
- **Clock Synchronization in MIMO**

Clock Synchronization in MIMO

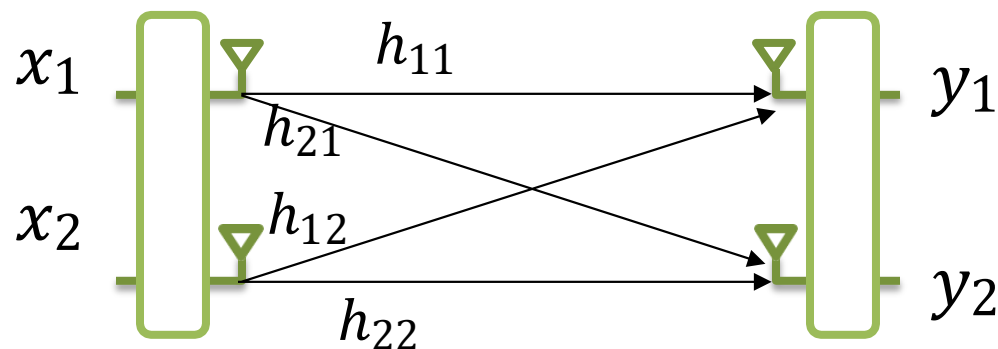


$$y_1 = h_{11}x_1e^{-j2\pi\Delta f_{11}t} + h_{12}x_2e^{-j2\pi\Delta f_{12}t}$$

$$y_2 = h_{21}x_1e^{-j2\pi\Delta f_{21}t} + h_{22}x_2e^{-j2\pi\Delta f_{22}t}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

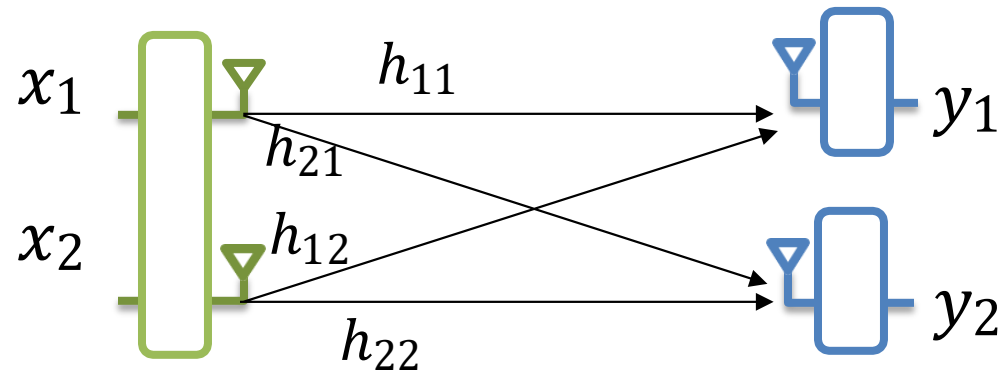


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

On chip MIMO: $\Delta f_{11} = \Delta f_{21} = \Delta f_{12} = \Delta f_{22} = \Delta f$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} e^{-j2\pi\Delta f t} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

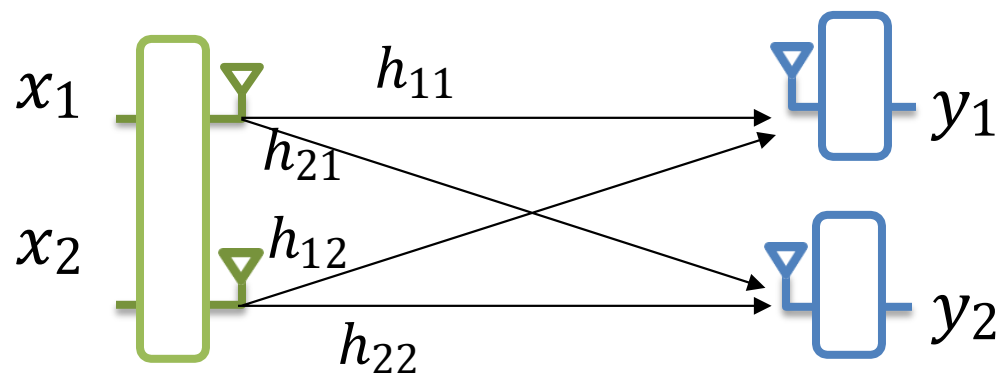


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_1t} & h_{12} e^{-j2\pi\Delta f_1t} \\ h_{21} e^{-j2\pi\Delta f_2t} & h_{22} e^{-j2\pi\Delta f_2t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

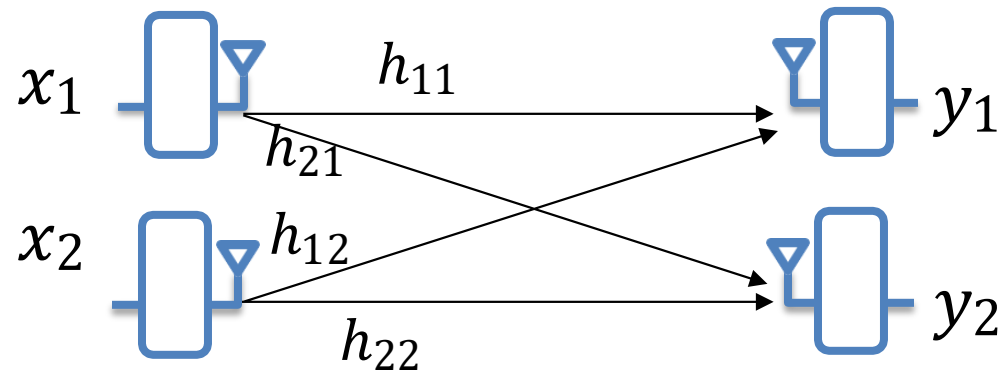


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e^{-j2\pi\Delta f_1 t} & 0 \\ 0 & e^{-j2\pi\Delta f_2 t} \end{bmatrix} \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Distributed MIMO: $\Delta f_{11} \neq \Delta f_{12} \neq \Delta f_{21} \neq \Delta f_{22}$